Special instructions for students

Unless otherwise stated the following parameters have the following definitions:

V_{DS} *: Drain Source Voltage.*

V_{GS} *: Gate Source Voltage.*

V_{TH} *: Threshold Voltage.*

U_t *: Thermal Voltage.*

n *: Weak Inversion Slope factor.*

g_m*: Transconductance.*

Solutions

1.

a. Compare Voltage and Current mode domains as regards to dynamic range, processing, signal distribution and speed.

- In voltage mode dynamic range is limited by supply voltage whereas is current mode dynamic range is limited by how much current can be sourced or sinked in a current mirror leading to larger dynamic range. [1 point]
- Current mode can lead to easier signal processing since addition of signals required just summing through Kirchoffs current law. [1 point]
- In voltage mode one wire can feed many nodes whereas in current mode one wire serves one node and current mirrors are needed to copy currents. [1 point]
- Current mode operation leads to higher speed of operation since capacitive nodes can be charged instantaneously. [1 point]

[4]

b) The circuit shown in Figure 1.1 has MOS transistors which are operating in the weak-inversion region of operation with a slope factor n = 1.23.

i) Show that the output current $I_{out} = I_1 - I_2$ of the circuit may be given by equation 1. Show all your steps in derivation including the derivation of currents I_1 and I_2.

$$I_{out} = I_{DC}\sinh\left(\frac{V_1 - V_2}{nU_t}\right) (1)$$

[Derivation from notes]

$$I_1 = \frac{I_{DC}}{2} e^{\left(\frac{V_+ - V_-}{nU_t}\right)} \qquad I_2 = \frac{I_{DC}}{2} e^{\left(\frac{-(V_+ - V_-)}{nU_t}\right)}$$ [1+1]

and

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$ [1]

Thus

$I_1 - I_2 = I_{DC}\sinh\left(\frac{V_1 - V_2}{nU_t}\right)$ [1]

[4]

ii) Derive the small signal transconductance and thus derive the transconductance efficiency of the circuit in Figure 1.1 stating all your assumptions. You may use the Taylor series expansion $coshx = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$ in your analysis.

Assuming small signals, we only take the first term of the Taylor Expansion [1]

$Transconductance = \frac{dI_{out}}{dV_{in}} = \frac{I_{DC}}{nU_t}\cosh\left(\frac{V_1 - V_2}{nU_t}\right) \approx \frac{I_{DC}}{nU_t}$ [2]

$Transconductance\ efficiency = 1/nU_t$ [1]

[4]

b. For the cascade of amplifiers shown in Figure 1.1, derive the output referred noise and explain what factors can be optimized to improve the signal to noise ratio.

Mos Noise: $$v_n^2 = \frac{K_w}{I_D^p}\cdot\Delta f + \frac{K_f}{A}\cdot\ln\left(\frac{f_h}{f_l}\right)$$

Distributed gain: $$G_i = \prod_{k=i}^{k=M} g_k$$

Solutions

Total output Noise: $$v_{no}^2 = \sum_{i=1}^{i=M} v_{ni}^2 G_i^2$$

Final Equation: $$v_{no}^2 = \sum_{i=1}^{i=M} v_{ni}^2 G_i^2 = \sum_{i=1}^{i=M} \left(n_i \frac{K_w}{(I_i / n_i)^p} \cdot \Delta f + n_i \frac{K_f}{(A_i / n_i)} \cdot \ln\left(\frac{f_h}{f_l}\right) \right) G_i^2$$ [6 points]

- Increasing Power and Area reduce the noise
- If we are willing to compute slowly (Δf and fh/fl are both small) then we can keep output noise low without consuming a lot of power. [2 points]

[8]

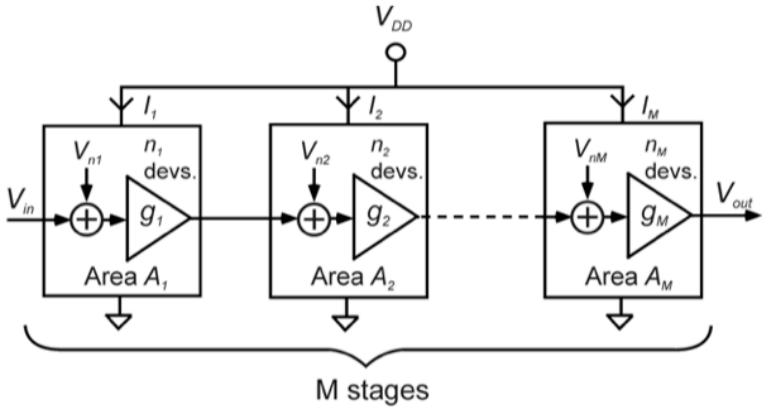


Figure 1.1: Cascade of amplifiers

Solutions

2.

a. State and derive the translinear principle (TLP) for a loop of MOS transistors working in weak inversion. Give all assumptions you make.

[bookwork]

In a closed loop containing an even number of forward biased junctions, arranged so that there are an equal number of clockwise facing and counterclockwise facing polarities, the product of the current densities in the clockwise direction is equal to the product of the current densities in the counterclockwise direction.

[2 points for definition]

Number of Clockwise Junctions (Voltage Rises)=Number of Anticlockwise Junctions (Voltage drops)

$$\sum_{n\in CW} V_n = \sum_{n\in CCW} V_n$$

$$V_n = nU_T \ln\left(\frac{I_n}{\lambda_n I_{DO}}\right)$$

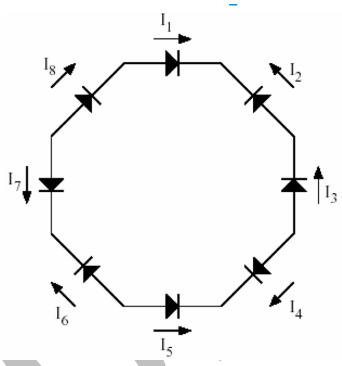
$$\sum_{n\in CW} nU_T \ln\left(\frac{I_n}{\lambda_n I_{DO}}\right) = \sum_{n\in CCW} nU_T \ln\left(\frac{I_n}{\lambda_n I_{DO}}\right)$$

$$\sum_{n\in CW} \ln\left(\frac{I_n}{\lambda_n I_{DO}}\right) = \sum_{n\in CCW} \ln\left(\frac{I_n}{\lambda_n I_{DO}}\right)$$

$$\prod_{n\in CW} \frac{I_n}{\lambda_n I_{DO}} = \prod_{n\in CCW} \frac{I_n}{\lambda_n I_{DO}}$$

$$\prod_{n\in CW} \frac{I_n}{\lambda_n} = I_{DO}^{CW-CCW} \prod_{n\in CCW} \frac{I_n}{\lambda_n}$$

$$\prod_{n\in CW} \frac{I_n}{\lambda_n} = \prod_{n\in CCW} \frac{I_n}{\lambda_n}$$



Assumptions: Temperature and saturation currents are all the same.

[3 points for derivation]

[5]

b. Show how short channel effects in MOS transistors operating in weak inversion, lead to multiplicative errors in output currents of translinear circuits.

- Short channel effects of MOS transistors induce an Error due to early effect/channel length modulation

$$I_{DS(wi)} = \lambda \cdot I_{D0} \exp\left(\frac{V_{GS}}{nU_t}\right)\left(1 - \exp\left(\frac{-V_{DS}}{U_t}\right)\right) = \lambda \cdot \gamma \cdot I_{D0} \exp\left(\frac{V_{GS}}{nU_t}\right)$$ [2]

- Appear like an area mismatch

$$\prod_{n\in CW} \frac{I_n}{\lambda_n \gamma_n} = \prod_{n\in CCW} \frac{I_n}{\lambda_n \gamma_n}$$ [2]

[4]

c. Figure 2.1 shows a translinear circuit.

Solutions

i. Derive the transfer function I_{out} of this circuit using the translinear principle. You may assume that currents I_1 and I_3 are input currents and I4 and I2 are a static bias'.

Translinear Loop: $I_{Q1}.I_{Q2}.I_{Q3}.I_{Q4}.I_{Q5}=I_{Q6}.I_{Q7}.I_{Q8}.I_{Q9}.I_{Q10}$ [Total 2 points]
Also $I_{Q1}=I_{Q2}=I_{out}-I_2$, $I_{Q4}=I_{Q5}=I_{out}$, $I_{Q6}=I_{Q7}=I_2$, $I_{Q10}=I_{Q9}=I_4$
Substituting yields:
$I_{out}=I_4\sqrt{(I_3/I_1)}$ [Total 3 points]

[5]

ii. Write down an expression for the output current I_{out} when I_{in}= Asin(ωt) and I_1 and I_3 are given as:

$$I_3 = \left|\frac{dI_{in}}{dt}\right|, \qquad I_1 = \left|\int I_{in}\, dt\right|$$

$I_3 = \left|\frac{dI_{in}}{d_t}\right| = A\omega cos(\omega t)$, [1]
$I_1 = |\int I_{in}\, dt| = \frac{A}{\omega}\cos(\omega t)$ [1]
$I_{out}=I_4$ x ω [2]

[4]

iii. What is the function of this circuit when the inputs are applied as in 2.c.ii ?
The circuit monitors the frequency of the input [Total 2 points]

[2]

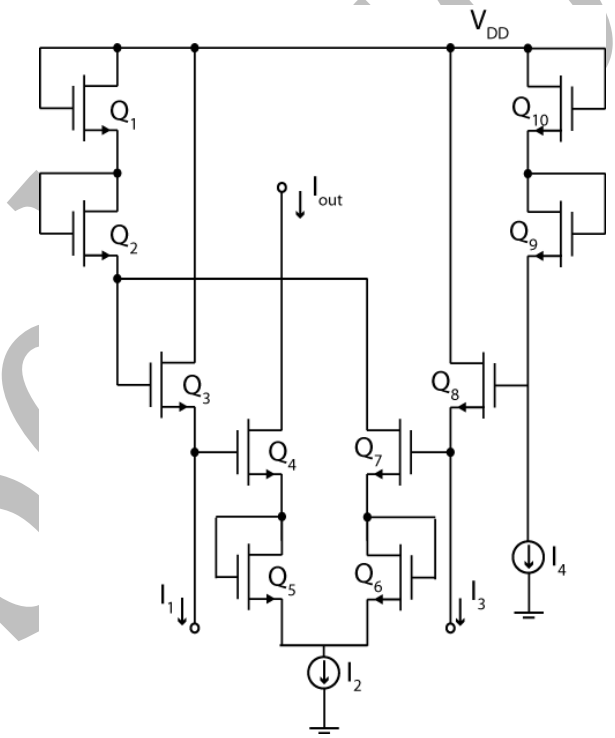


Figure 2.1

3. a) Figure 3.1 shows a block diagram of a current-mode companding integrator, with an input X and an output Y. f() and g() represent non-linear functions which compress and expand the signal such that the input-output relationship of the integrator is linear and given by:

$$Y = \tau \int X \cdot dt$$

Given that the expansive function is defined as $I_{out} = g(V_c) = I_0 \sinh(V_c/nU_t)$, derive the function f(x) such that linear integration is achieved. All constants should be grouped to represent a current I_1, with the function f(x) composed of just I_{in}, I_1 and I_{cosh}, whereby $I_{cosh} = I_0 \cosh(V_c/nU_t)$.

New derivation:

$$Y = I_{out} = I_0 \sinh(V_c / nU_t) = g(V_C)$$

$$\frac{d(g(V_c))}{dV_C} = \frac{dY}{dV_C} = \frac{dI_{out}}{dV_C} = \frac{I_0}{nU_t}\cosh(V_C / nU_t) = \frac{I_{Cosh}}{nU_t}$$

$$f(x) = I_C = X\tau C\left(\frac{dY}{dV_C}\right)^{-1} = X\tau C\frac{nU_t}{I_{\cosh}} = X\frac{I_1}{I_{\cosh}} \qquad X = I_{in}, \quad I_1 = \tau \cdot CnU_t$$

$$I_C = f(I_{in}) = \frac{I_{in}I_1}{I_{\cosh}}$$

[6]

b) The transfer function of a second order topology has been decomposed into the following state-space equations:

$$\dot{X}_1 = -\left(\frac{\omega_0}{2Q}\right)X_1 - \omega_0\left(1 - \frac{1}{4Q^2}\right)X_2 + \omega_0 U$$

$$\dot{X}_2 = -\left(\frac{\omega_0}{2Q}\right)X_2 + \omega_0 X_1$$

$$Y_1 = X_1$$

$$Y_2 = X_2$$

whereby Y_1 and Y_2 are the outputs and X_1 and X_2 are the state-variables and U is the input.

i) Show that the output Y_1 can be used to implement a second order lowpass transfer function and the output Y_2 can be used to implement a "two-pole one-zero" second order transfer function.

New derivation:

$$\left[s + \frac{\omega_0}{2Q}\right]X_1(s) - \omega_0\left(1 - \frac{1}{4Q^2}\right)X_2(s) = \omega_0 U$$

$$-\omega_0 X_1(s) + \left[s + \frac{\omega_0}{2Q}\right]X_2(s) = \omega_0 U(s)$$

Substituting for $X_2(s)$ (2 points)

$$\frac{Y_1(s)}{U(s)} = \frac{X_1(s)}{U(s)} = \frac{\omega_0^2}{s^2 + \left(\frac{\omega_0}{Q}\right)s + \omega_0^2}, \textit{low pass response}$$

Solutions

Substituting for $X_1(s)$ (2 points)

$$\frac{Y_2(s)}{U(s)} = \frac{X_2(s)}{U(s)} = \frac{\omega_0 \left(s + \frac{\omega_0}{2Q}\right)}{s^2 + \left(\frac{\omega_0}{Q}\right)s + \omega_0^2}, \textit{two pole, one zero response}$$

[4]

ii) By using the mappings below, show how these state space equations can be mapped to non-linear log-domain design equations. State any assumptions you make.

$$X_1 = I_1 \exp\left(\frac{V_1}{nU_t}\right) \qquad X_2 = I_2 \exp\left(\frac{V_2}{nU_t}\right) \qquad U = I_U \exp\left(\frac{V_U}{nU_t}\right)$$

New derivation from Theory:

$$\frac{I_1}{nU_t}\exp\left(\frac{V_1}{nU_t}\right)\dot{V}_1 = -\left(\frac{\omega_0}{2Q}\right)I_1 \exp\left(\frac{V_1}{nU_t}\right) - \omega_0\left(1 - \frac{1}{4Q^2}\right)I_2 \exp\left(\frac{V_2}{nU_t}\right) + \omega_0 I_u \exp\left(\frac{V_u}{nU_t}\right)$$
$$\frac{I_2}{nU_t}\exp\left(\frac{V_2}{nU_t}\right)\dot{V}_2 = -\left(\frac{\omega_0}{2Q}\right)I_2 \exp\left(\frac{V_2}{nU_t}\right) + \omega_0 I_1 \exp\left(\frac{V_1}{nU_t}\right)$$

$$C\dot{V}_1 = -CnU_t\left(\frac{\omega_0}{2Q}\right) - CnU_t\omega_0\left(1 - \frac{1}{4Q^2}\right)\frac{I_2}{I_1}\exp\left(\frac{V_2 - V_1}{nU_t}\right) + CnU_t\omega_0\frac{I_u}{I_1}\exp\left(\frac{V_u - V_1}{nU_t}\right)$$
$$C\dot{V}_2 = -CnU_t\left(\frac{\omega_0}{2Q}\right) + CnU_t\omega_0\frac{I_1}{I_2}\exp\left(\frac{V_1 - V_2}{nU_t}\right)$$

Let C1=C2=C and define: $I_\omega = CnU_t\omega_0$
A suitable mapping:

$$I_1 = I_2 = I_\omega$$
$$I_U = I_0$$

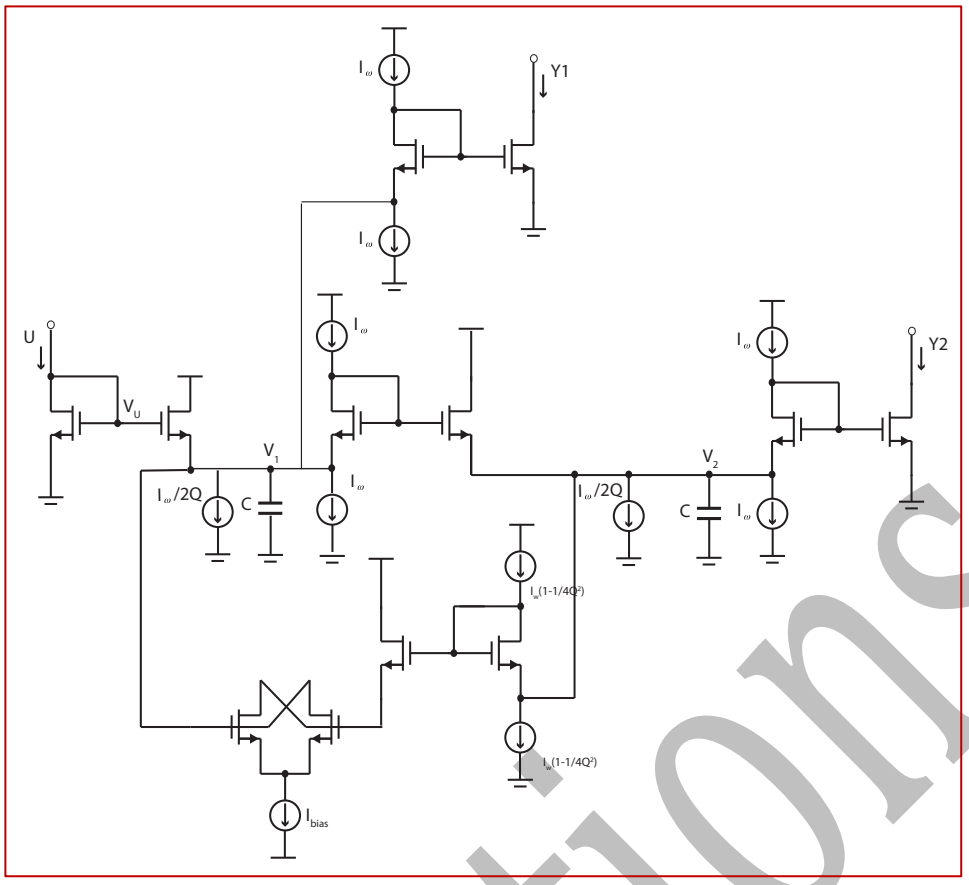
Final Equations:

$$C\dot{V}_1 = -\left(\frac{I_\omega}{2Q}\right) - I_\omega\left(1 - \frac{1}{4Q^2}\right)\exp\left(\frac{V_2 - V_1}{nU_t}\right) + I_o \exp\left(\frac{V_u - V_1}{nU_t}\right)$$
$$C\dot{V}_2 = -\left(\frac{I_\omega}{2Q}\right) + I_\omega \exp\left(\frac{V_1 - V_2}{nU_t}\right)$$
$$Y_1 = I_\omega \exp\left(\frac{V_1}{nU_t}\right)$$
$$Y_2 = I_\omega \exp\left(\frac{V_2}{nU_t}\right)$$

[6]

iii) With these log-domain design equations, sketch a schematic of the final log domain topology which realises the two outputs Y_1 and Y_2 using weak inversion MOS transistors.

Solutions



[10]

Solutions

4.

a) Figure 4.1 shows a conventional second-generation positive current conveyor, CCII+.

(i) Explain it's operation principle and describe how impedances at each of its three ports, X, Y, Z, differ from a standard operational amplifier.

[bookwork]

- Current-voltage characteristics
  $V_X=V_Y$, $I_Z=\pm I_X$, $I_Y=0$
  $Z_Y \rightarrow \infty$, $Z_X \rightarrow 0$, $Z_Z \rightarrow \infty$ [1]
- Voltage-follower between Y-input and X-output
- Current-follower between X-input and Z-output [1]

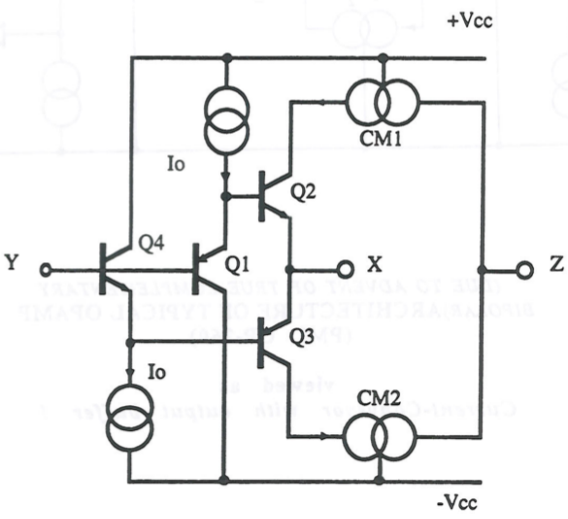
Plus a comparison with Op-amp terminals [1]

[4]

(ii) Draw the circuit of a transistor implementation of a bi-directional CCII+ which does not use any opamps.

Application of taught circuits

Schematic:

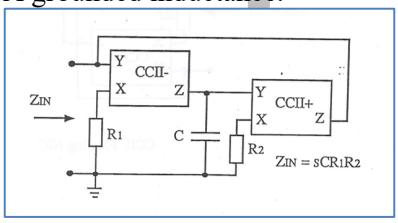


Internal Schematic of CCII01

[4]

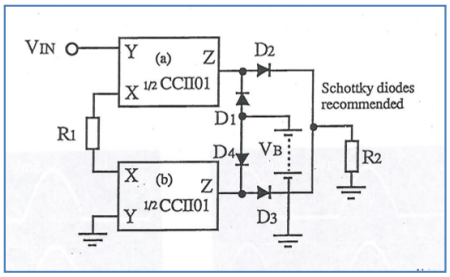
Using current conveyors design circuits which implement:

A grounded inductance.



[2]

A full wave voltage mode rectifier.



[2]

[4]

c) Derive the output current, I_{out} for the switch current algorithmic cell given in Figure 5.2 showing how this eliminates any error caused due to the switches.

Derivation from Theory:

Operation

Assuming a charge injection δ

- Period 1: I_{ds1}-J=δ1
- Period 2: I_{ds2}-J=-δ1+δ2
- Period 3: I_{ds3}-J=I_{in}+δ1-δ2+δ3
- Period 4: I_{ds1}-J=I_{in}+δ1-I_{ds3}=+δ2-δ3
- Period 5: I_{ds2}-J=δ2-δ2+δ3=δ3
- Period 6: I_{ds3}=I_{in}+δ3-δ3=I_{in}
- Period 7: I_{out}=I_{in} !

[4]

5. This question is related to switched capacitor circuits.

(a) State four non-idealities of using MOSFETs as switches in switched capacitor circuits and write the equation for ΔV_{total} considering all four factors. (4)

1. Charge Injection, 2. Clock feedthrough, 3. Thermal noise (KT/C) 4. Leakage current (2)

$$\Delta V_{total} = \frac{q_{inj}}{C_H} + \frac{WC_{ov}}{WC_{ov} + C_H} + \sqrt{\frac{KT}{C_H}} + \frac{I_{leak} \times T_H}{C_H}$$

(2)

(b) For the most dominant non-ideality factor, explain using equations and diagram how the output deviates from the ideal input, considering body effect. (4)

The total channel charge to form the channel for a MOSFET transistor

$$Q_{ch} = WLC_{ox}(V_{DD} - V_{in} - V_{TH})$$

$$V_{out} = V_{in} - \Delta V = V_{in} - \frac{Q_{ch}}{C_H}$$

$$= V_{in}\left(1 + \frac{WLC_{ox}}{C_H}\right) - \frac{WLC_{ox}}{C_H}(V_{DD} - V_{TH})$$

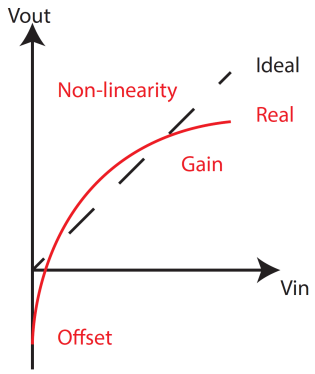
Body effect is given by

$$V_{TH} = V_{TH0} + \gamma(\sqrt{2\phi_B + V_{SB}} - \sqrt{2\phi_B}) \text{ and } V_{SB} = V_{in}$$

$$V_{out} = V_{in}\left(1 + \frac{WLC_{ox}}{C_H}\right) + \gamma\frac{WLC_{ox}}{C_H}\sqrt{2\phi_B + V_{in}} - \frac{WLC_{ox}}{C_H}(V_{DD} - V_{TH0} + \sqrt{2\phi_B})$$

(2)

Solutions



NOTE: End point and bending direction of the red curve are not important, demonstrating the three factors on the plot is sufficient.

(2)

(c) Define a Figure of Merit (FoM) that considers the trade-off in speed and precision, with only first order effects. Based on the result, explain how you would design the size of the switch transistor using an n-MOSFET in a 180nm CMOS process to achieve a good FoM. (3)

$$FoM = \frac{1}{\tau \times \Delta V} = \frac{1}{R_{on} \times C_H \times \frac{Q_{CH}}{C_H}}$$

$$= \frac{1}{\frac{1}{\mu_n C_{ox} \frac{W}{L}(V_{DD} - V_{in} - V_{TH})} \times WLC_{ox}(V_{DD} - V_{in} - V_{TH})}$$

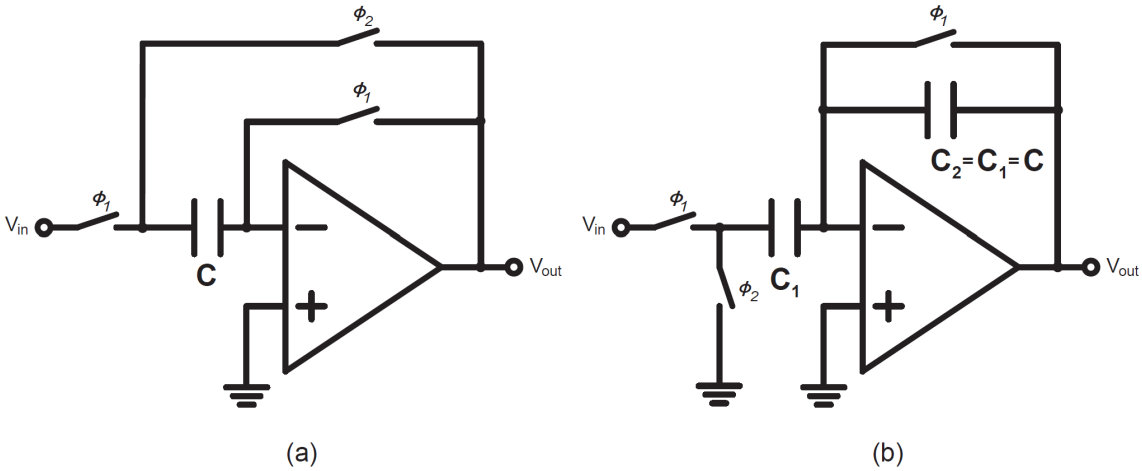
$$= \frac{\mu_n}{L^2}$$

(2)

Use minimum length NMOS: L = 180nm.

(1)

(d) State the trade-offs with respect to gain, precision and speed between Fig 5.1 (a) and Fig 5.1 (b).(3)



(e) Fig 5.1

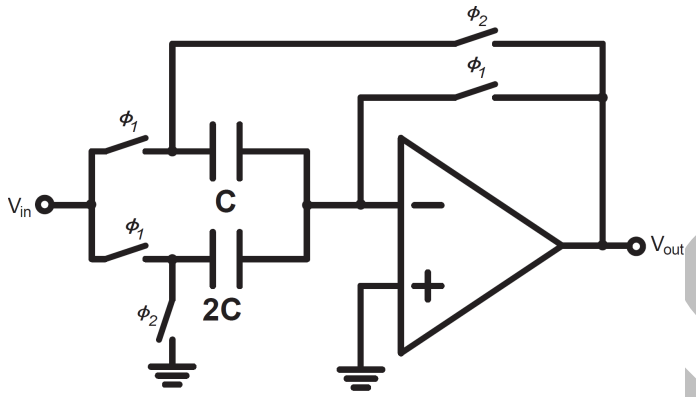
Fig 5.1 (a) is a unity gain buffer / sampler with a gain of 1, feedback factor of 1 (assuming the opamp has 0 input capacitance.)
Fig 5.1 (b) is a non-inverting amplifier with a gain of 1, feedback factor of 0.5.

Solutions

The open loop gain of the amplifier: A_{OL} is attenuated by the feedback factor of each circuit respectively. Therefore the precision and speed of Fig 5.1 (b) is worse than that of Fig 5.1(a) provided that they achieve the same gain.

(3)

(e) Based on the conclusion of Q5(d), design a precision multiply-by-3 circuit. Explain its operating principle and describe how it achieves better precision compared to the equivalent of the circuit in Fig 5.1 (b) when C1 = 3 x C2. (4)



(2)

$\phi 1$: $Q_t = V_{in} \times (C + 2C)$
$\phi 2$: $Q_t = V_{out} \times C$
Due to charge conservation $Q_{t-\phi 1} = Q_{t-\phi 2}$, thus $V_{out} = 3 \times V_{in}$
The feedback factor of this circuit in $\phi 2$ is $\frac{1}{3}$
For Fig 5.1 (b), if C1 = 3C2, the feedback factor is $\frac{1}{4}$
Gain error is defined by $1 + \frac{1}{A \times fb}$
Therefore to achieve the same gain of 3, the equivalent circuit of Fig 5.1(b) has more gain error compared to the precision x3 circuit.

(2)

(f) Redraw the circuit in Fig 5.1(b) with polarised capacitors in CMOS processes, as modelled in Fig 5.2. Explain your choice of polarisation. (2)

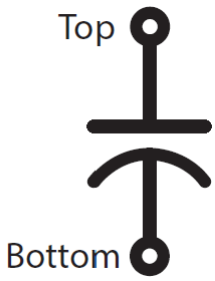
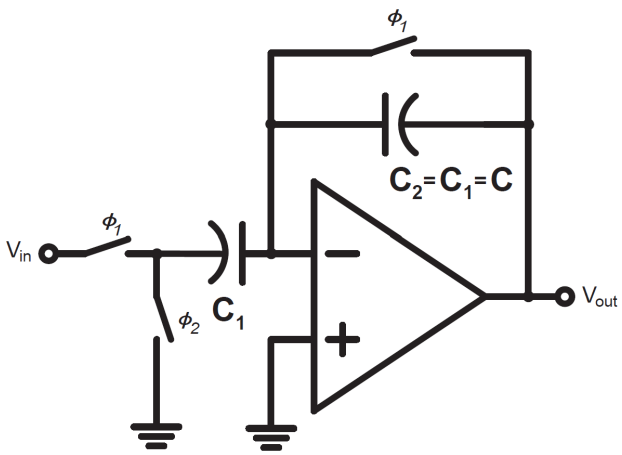


Fig 5.2

Redraw the circuit of Fig 5.1(b) with replacement of polarised capacitor, and connect the bottom plate away from the virtual ground node.

Solutions



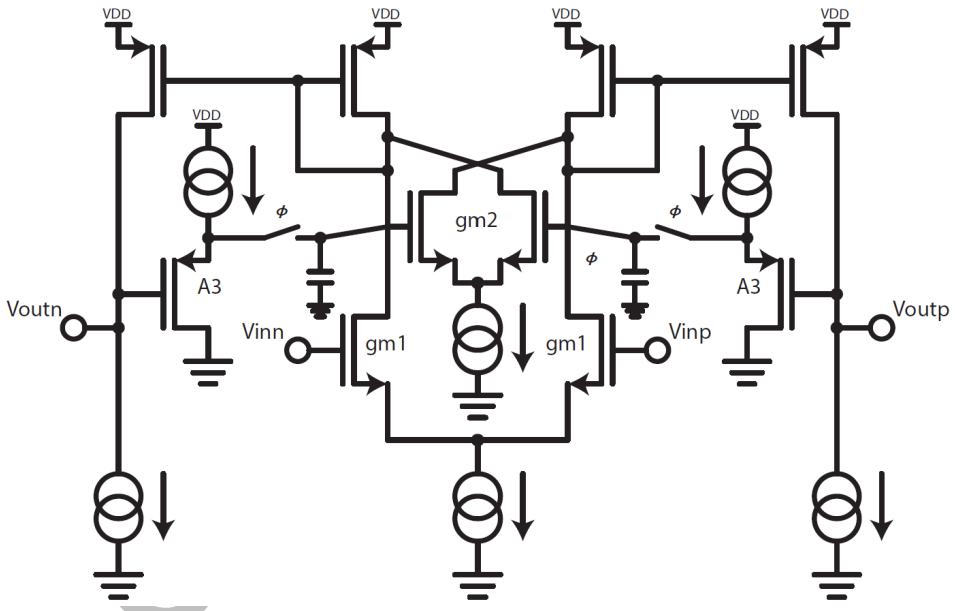
(1)

Bottom Plate Sampling, to reduce the parasitics at the virtual ground node of the amplifier.

(1)

6. This question is related to a sensor instrumentation system using switch capacitor techniques.

a) Design a closed-loop auto-zeroing circuit using auxiliary compensation for a fully differential (differential input and differential output) 2-stage amplifier, and state the residual offset seen at the input of the main amplifier.



(3)

$$V_{os-res} = V_{os}(1 - \frac{A_2A_3}{1 + A_2A_3}) \approx \frac{V_{os}}{A_2A_3}$$

(1)

[4]

b) State the function of the two circuits shown in Fig 6.1 and state the main benefit of the scheme shown in Fig 6.1(b).

They are both switched capacitor non-inverting amplifier, however Fig 6.1(b) uses anticipatory compensation (prediction) to relax the bandwidth requirement from the

opamp, so that any input change during $\phi 1$ is memorised and pre-amplified by the circuit. (2)

[2]

c) Analyse the circuit in shown Fig 6.2 deriving its output voltage at each phase of operation and state its functionality.

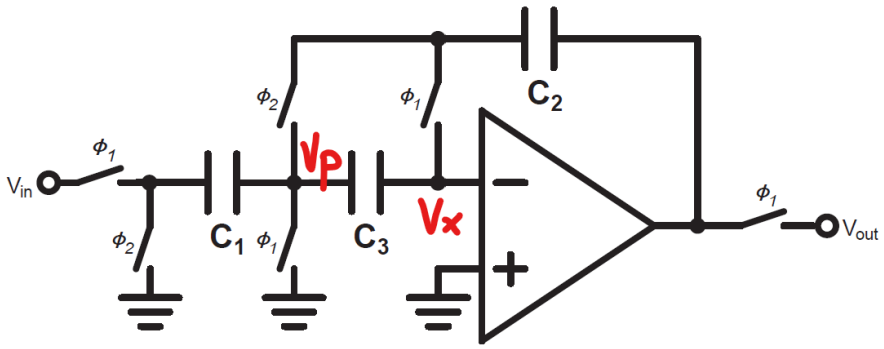


Fig 6.2

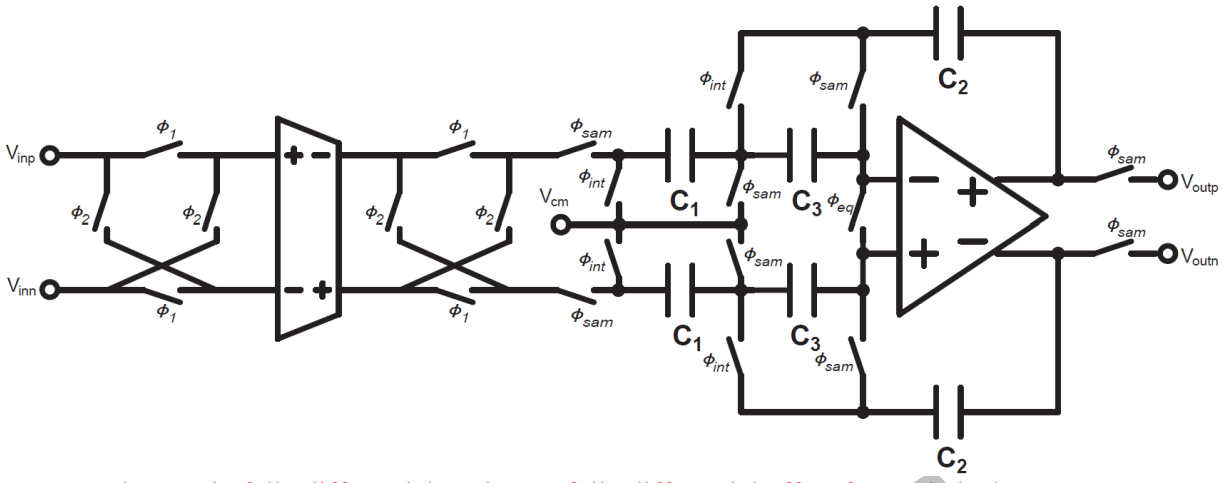
$\phi 1$: $V_{c1} = V_{in}, V_{c2} = V_{out} - V_{os}, V_{c3} = V_{os}, V_p = 0$
$\phi 2$: At the end of charge transformation: $V_x = V_{os}, V_p = 0$.
Thus: $-V_{in} \times C_1 = \Delta V_{out} \times C_2, \Delta V_{out} = -V_{in} \times \frac{C_1}{C_2}$
Next $\phi 1$, $V_{out-next} = V_{out-\phi 2} = V_{out-previous} - V_{in} \times \frac{C_1}{C_2}$
It's an offset free switched capacitor integrator operating with correlated double sampling technique.

[4]

d) (i) Fig. 6.3 shows a typical analogue signal processing chain that continuously amplifies an output signal from a differentially configured sensor pair to a differential input ADC. You are responsible for designing a signal processing pipeline using the following criteria:

- The sensor's output has a frequency range of 1–10 Hz, and a signal magnitude of 10 mV. Each op-amp has a 1/f noise amplitude of 10 mV at 10 Hz, and a corner frequency of 1 kHz (you don't need to show the circuit inside the opamp).
- You should design the circuit so that the sensor's output is not buried by the input-referred noise of your circuit.
- You must use the circuit in Fig 6.2 as one of your blocks in the pipeline.

Solutions



Chopper in fully differential, and use a fully differential offset-free switched capacitor integrator (modified Fig 6.2) as a low pass filter to filter out the high frequency noise.

NOTE: Any similar and reasonable answer can be considered.

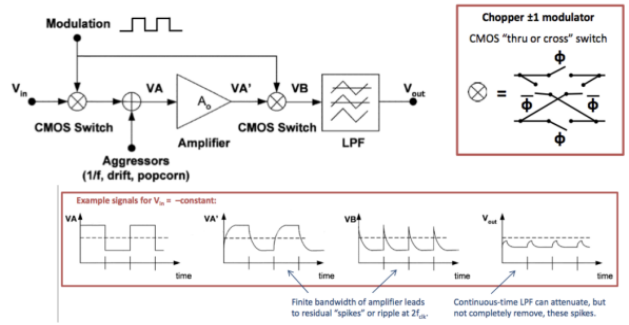
(4)

[4]

(ii) Explain the method used to achieve a good noise performance in your sensing system. You may use frequency spectrum plots to aid your description.

The Chopper Stabilisation Technique

- Upmodulate input signal to frequency above amplifier's 1/*f* noise corner, amplify ac signal, then downmodulate to original frequencies.
- Drift, offset, and 1/*f* noise from amplifier pass only through the second modulator, so they are upmodulated and removed by the low-pass filter.



Frequency Spectrum plots:

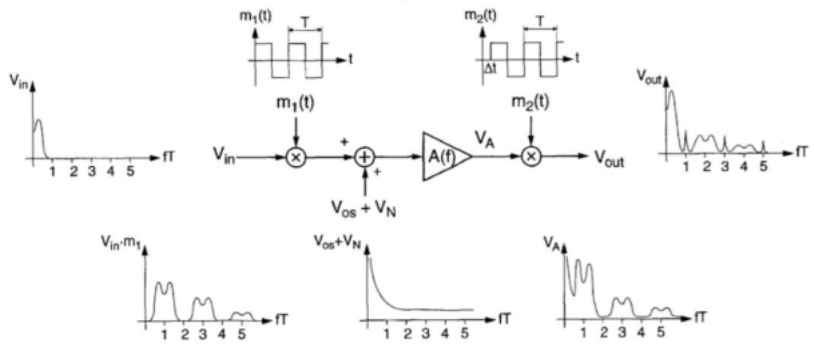


Fig. 9. The chopper amplification principle [19].

(4)

[4]

(iii) State the effect at the input of the ADC due to a finite bandwidth of the main amplifier.

Finite bandwidth leads to an amplitude reduction.

The Chopper Stabilisation Technique

- Effects of finite bandwidth of op-amp.
- A square wave is rich in harmonics and not all will go through as finite bandwidth means they will get low-pass filtered.
- Assume output will be only first harmonic, i.e a sinewave of amplitude $(4/\pi).A_0.V_{in}$.
- After demodulation the output will be a rectified version.
- Lowpass filtering gives $(8/\pi^2).A_0.V_{in} \cong 0.8.A_0$
- Finite bandwidth leads to an amplitude reduction.

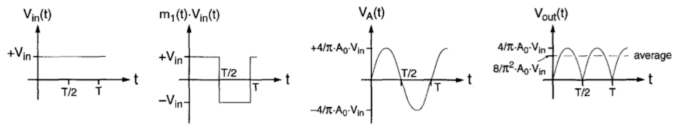


Fig. 10. Waveforms appearing along the chopper amplifier for a dc input and an amplifier

(2)

[2]

Solutions